Power Series

Section 9.8

3-5-26

A) Previously...

Ex: $\sum_{1}^{\infty} \frac{5^n}{n!} = 5 + \frac{5^2}{2!} + \frac{5^3}{3!} + \ldots$

B) What makes Power Series converge?

	Radius
1) $-\infty < x < \infty$	$R = \infty$
2) $x = c$	$R = 0$
3) $a < x < b$	$R = \#$

Ex:

$$\sum_{0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\lim_{n\to\infty} \left| \frac{(-1)^{n+1} x^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{(-1)^n x^{n+1}} \right|$$

$$\lim_{n\to\infty} \left| \frac{x^2}{(2n+3)(2n+2)} \right| = 0 < 1$$

$$\boxed{\begin{matrix} -\infty < x < \infty \\ R = \infty \end{matrix}}$$

Ex: $\sum_{1}^{\infty} n!(x-3)^n$

$$\lim_{n\to\infty} \left| \frac{(n+1)!(x-3)^{n+1}}{n!(x-3)^n} \right|$$

$$\lim_{n\to\infty} \left| (n+1)(x-3) \right| = \infty > 1$$

$$\boxed{\begin{matrix} \text{Converges at } x=3 \\ R=0 \end{matrix}}$$

Ex: $\sum_{1}^{\infty} \frac{(-1)^{n+1}(x-5)^n}{n \cdot 2^n}$

$$\lim_{n\to\infty} \left| \frac{(x-5)^{n+1}}{(n+1)2^{n+1}} \cdot \frac{n \cdot 2^n}{(x-5)^n} \right|$$

$$\lim_{n\to\infty} \left| \frac{(x-5)n}{2(n+1)} \right| = \frac{|x-5|}{2} < 1$$

$|x-5| < 2$ $\boxed{R=2}$

$3 < x < 7$

$x = 7$ $\sum_{1}^{\infty} \frac{(-1)^{n+1} \cancel{2^n}}{n \cdot \cancel{2^n}} = \sum_{1}^{\infty} \frac{(-1)^{n+1}}{n}$

alt, $\lim_{n\to\infty} \frac{1}{n} = 0$, $\frac{1}{n+1} < \frac{1}{n}$

$\boxed{\text{converges by alternating series}}$

$x = 3$ $\sum_{1}^{\infty} \frac{(-1)^{n+1}(-2)^n}{2^n \cdot n} = \sum_{1}^{\infty} \frac{(-1)^{n+1}(-1)^n \cdot 2^n}{n \cdot 2^n}$

$$\sum_{1}^{\infty} \frac{(-1)^{2n+1}}{n} = \sum_{1}^{\infty} \frac{-1}{n}$$

diverge by harmonic

$3 < x \leq 7$ $R = 2$

Ex: $\sum_{0}^{\infty} 3(x-2)^n$

geometric

$|x-2| < 1$ $r = 1$

$\boxed{\begin{matrix} 1 < x < 3 \\ r = 1 \end{matrix}}$ no need to test endpoints